

TOPOLOGY

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Abstract. This is a lecture note on Topology and Complex Analysis (MA21206) that I gave at IIT Kharagpur.

There might be a typographical or unintentional error in the note. If you find one, kindly bring it to my notice.

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1. PRELIMINARIES

Definition 1.1. A set is a collection of well-defined, distinct objects. Object that belongs to the set are called elements.

Examples 1.2. The following are some examples of a set.

- (1) All the students of IIT KGP.
- (2) $\{a, b, c\}$.
- (3) The set of all natural numbers $= \mathbb{N} = \{1, 2, 3, \dots\}$.
- (4) The set of all integers $= \mathbb{Z} = \{0, +1, -1, +2, -2, \dots\}$.
- (5) The set of all rational numbers $= \mathbb{Q} = \{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\}$.
- (6) The set of all real numbers $= \mathbb{R}$ (construction of this will be a project).
- (7) The set of all complex numbers $= \mathbb{C} = \{a + ib : a, b \in \mathbb{R}, i = \sqrt{-1}\}$.

Definition 1.3. Let A and B be sets. A *function* $f : A \rightarrow B$ is a rule that assigns to each element $a \in A$ exactly one element $b \in B$.

- A is called the *domain* of f .
- B is called the *codomain* of f .
- The assigned element b is denoted by $f(a)$.

Equivalently, a function is a subset $f \subseteq A \times B$ such that for every $a \in A$ there exists a *unique* $b \in B$ with $(a, b) \in f$.

Definition 1.4. (1) A function $f : A \rightarrow B$ is called *injective* (or *one-to-one*) if

$$f(a_1) = f(a_2) \implies a_1 = a_2 \quad \text{for all } a_1, a_2 \in A.$$

- (2) A function $f : A \rightarrow B$ is called *surjective* (or *onto*) if for every $b \in B$ there exists at least one $a \in A$ such that

$$f(a) = b.$$

- (3) A function $f : A \rightarrow B$ is called *bijective* if it is both injective and surjective. Equivalently, f is bijective if for every $b \in B$ there exists a *unique* $a \in A$ such that

$$f(a) = b.$$

¹If you find any errors or typos, please let me know AT parvez@maths.iitkgp.ac.in

Definition 1.5. Let A and B be sets. A *relation* ρ from A to B is any subset of the Cartesian product $A \times B$. If $(a, b) \in \rho$, we say that a is *related* to b and write $a \sim b$.

In particular, a relation ρ on a set A is a subset $\rho \subseteq A \times A$.

Definition 1.6. Let A be a set. A relation ρ on A is called an *equivalence relation* if it satisfies the following properties for all $a, b, c \in A$:

- **Reflexive:** $(a, a) \in \rho$.
- **Symmetric:** If $(a, b) \in \rho$, then $(b, a) \in \rho$.
- **Transitive:** If $(a, b) \in \rho$ and $(b, c) \in \rho$, then $(a, c) \in \rho$.

For any $a \in A$, the *equivalence class* of a , denoted by $[a]$, is defined by

$$[a] = \{ x \in A \mid (x, a) \in \rho \}.$$

Examples 1.7. Let $A = \mathbb{Z}$. Define a relation ρ on A by

$$(a, b) \in \rho \quad \text{if and only if} \quad a - b \text{ is divisible by } 5.$$

That is,

$$(a, b) \in \rho \iff a - b = 5k \text{ for some } k \in \mathbb{Z}.$$

This relation is an equivalence relation:

- **Reflexive:** $a - a = 0 = 5 \cdot 0$, so $(a, a) \in \rho$.
- **Symmetric:** If $a - b = 5k$, then $b - a = -5k$, so $(a, b) \in \rho$.
- **Transitive:** If $a - b = 5k$ and $b - c = 5m$, then $a - c = 5(k + m)$, so $(a, c) \in \rho$.

Question 1.8. Find the equivalence class of the above relation.

Definition 1.9. A set B is said to be *finite* if there exists a bijection from B to $\{1, \dots, n\}$ where $n \in \mathbb{N}$ or $B = \emptyset$.

Excercise 1.10. (1) *Arbitrary intersection of finite sets is finite.*
 (2) *Finite union of finite sets is finite.*

Definition 1.11. A set B is said to be countable if B is finite or if there exists a bijection from $\mathbb{N}$ to B .

2. TOPOLOGY

Definition 2.1. Let X be a set. A collection $\mathcal{T} \subseteq \mathcal{P}(X)$ (the power set of X) is called a *topology* on X if it satisfies the following conditions:

- $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$.
- The union of any collection of sets in $\mathcal{T}$ belongs to $\mathcal{T}$.
- The intersection of any finite number of sets in $\mathcal{T}$ belongs to $\mathcal{T}$.

The pair $(X, \mathcal{T})$ is called a *topological space*. The elements of $\mathcal{T}$ are called open sets of X .

Examples 2.2. (1) Let X be a set. The collection

$$\mathcal{T} = \{ \emptyset, X \}$$

is a topology on X , called the *trivial topology* (or *indiscrete topology*).

(2) Let X be a set. The collection

$$\mathcal{T} = \mathcal{P}(X)$$

consisting of all subsets of X is a topology on X , called the *discrete topology*. It is denoted by $\mathcal{T}_D$.

(3) Let $X = \{a, b, c\}$. Consider $\mathcal{T}_1 = \{\emptyset, \{a\}, X\}$ and $\mathcal{T}_2 = \{\emptyset, \{a\}, \{b\}, X\}$ are two subsets of $\mathcal{P}(X)$.

Claim: $\mathcal{T}_1$ is a topology and $\mathcal{T}_2$ is not a topology. (verify!!)

(4) Let X be a set. Define

$$\mathcal{T} = \{\emptyset\} \cup \{U \subseteq X \mid X \setminus U \text{ is finite}\}.$$

Then $\mathcal{T}$ is a topology on X , called the *cofinite topology*. It is denoted by $\mathcal{T}_{cf}$.

Verification.

- **Axiom 1:** Clearly, $\emptyset \in \mathcal{T}$. Also,

$$X \setminus X = \emptyset,$$

which is finite, hence $X \in \mathcal{T}$.

- **Axiom 2:** Let $\{U_\alpha\}_{\alpha \in I}$ be a family of sets in $\mathcal{T}$. If all $U_\alpha = \emptyset$, then $\bigcup_{\alpha \in I} U_\alpha = \emptyset \in \mathcal{T}$. Otherwise,

$$X \setminus \bigcup_{\alpha \in I} U_\alpha = \bigcap_{\alpha \in I} (X \setminus U_\alpha).$$

Each $X \setminus U_\alpha$ is finite, and an intersection of finite sets is finite. Hence $\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T}$.

- **Axiom 3:** Let $U_1, U_2, \dots, U_n \in \mathcal{T}$. Then

$$X \setminus \bigcap_{i=1}^n U_i = \bigcup_{i=1}^n (X \setminus U_i).$$

A finite union of finite sets is finite, so $\bigcap_{i=1}^n U_i \in \mathcal{T}$.

Thus, $\mathcal{T}$ satisfies all the axioms of a topology on X .

Exercise 2.3. (1) How many topology one can define on $X = \{a, b, c\}$? List all of them.

(2) Think about all the discussed topologies when $X = \mathbb{R}$ and $X = \mathbb{N}$.

Exercise 2.4. Prove that if X is finite then $\mathcal{T}_D = \mathcal{T}_{cf}$.

Exercise 2.5. Let X be any set. Prove that $\mathcal{T}_{cc} = \{U \subseteq X \mid X \setminus U \text{ is countable}\} \cup \{\emptyset\}$ is topology on X .

2.1. Basis of a topology.

Definition 2.6. X is any set. A basis for a topology on X is a collection $\mathcal{B}$ of X such that

- (1) for each $x \in X$, there exists $B \in \mathcal{B}$ such that $x \in B$.
- (2) if $x \in B_1 \cap B_2$, where $B_1, B_2 \in \mathcal{B}$, then there exists B_3 such that $x \in B_3 \subseteq B_1 \cap B_2$.

The elements of $\mathcal{B}$ are called basic elements.

Now we define the topology $\mathcal{T}$ generated by $\mathcal{B}$, where

$$\mathcal{T} = \{U \subseteq X \mid \text{for each } x \in U, \text{ there is a basis element } x \in B \text{ and } B \subseteq U\}$$

Claim: $\mathcal{T}$ is topology on X .

Proof of claim:

- **Axiom 1:** $\emptyset \subseteq X$, there is nothing to check. $X \in \mathcal{T}$ because for each $x \in X$ we have $B \in \mathcal{B}$ such that $x \in B \subseteq X$ (By Axiom 1 of the definition of Basis).
- **Axiom 2:** Let $\{U_\alpha\}_{\alpha \in I}$ be a family of sets in $\mathcal{T}$. Let $U = \bigcup_{\alpha \in I} U_\alpha$. We want to prove $U \in \mathcal{T}$. It is clear that $U \subseteq X$ because each $U_\alpha \subseteq X$. Let $x \in U = \bigcup_{\alpha \in I} U_\alpha \implies \exists \alpha \in I$ such that $x \in U_\alpha$. Note that $U_\alpha \in \mathcal{T}$, So $\exists B \in \mathcal{B}$ such that $x \in B$ and $B \subseteq U_\alpha \subset U$.

- **Axiom 3:** Let $U_1, \dots, U_n \in \mathcal{T}$. We want to prove $W = \cap_{i=1}^n U_i \in \mathcal{T}$. It is clear that $W \subseteq X$ as each $U_i \subseteq X$. We will prove this by induction on n . Here $n = 1$ case is clear. Assume we know the statement is true for $n = m$, then we want to prove it for $n = m + 1$. Let $W = \cap_{i=1}^m U_i \cap U_{m+1} = W_0 \cap U_{m+1}$. By induction $W_0 \in \mathcal{T}$ i.e. for each $x \in W_0$ $\exists B_0$ such that $x \in B_0 \subseteq W_0$. Also $U_{m+1} \in \mathcal{T}$, $\exists B_1$ such that $x \in B_1 \subseteq U_{m+1}$. By Axiom 2 of the definition of basis, $\exists B_2 \subseteq B_0 \cap B_1 \subset W_0 \cap U_{m+1} = W$. Hence $W \in \mathcal{T}$.

Exercise 2.7. Let $X = \mathbb{R}$ and $\mathcal{B} = \{(a, b) | a < b, a, b \in \mathbb{R}\}$. Prove that $\mathcal{B}$ is a basis for a topology.

Definition 2.8. X is any set. let $\mathcal{T}$ and $\mathcal{T}'$ be two topologies on X . If $\mathcal{T} \subseteq \mathcal{T}'$ then, we say $\mathcal{T}'$ is a finer topology than $\mathcal{T}$ or $\mathcal{T}$ is a coarser topology than $\mathcal{T}'$.

$\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3$ are topology on X . If there is a containment relation between topologies, then we say they are comparable; otherwise, they are not comparable.

Exercise 2.9. Provide the example of the finest topology and the example of the coarsest topology on a set X .

Lemma 2.10. Let X be a set and let $\mathcal{B}$ be a basis for a topology $\mathcal{T}$ on X . Then $\mathcal{T}$ is equal to the collection of all unions of elements of $\mathcal{B}$.

Proof. Let $\mathcal{C}$ = the collection of all unions of elements of $\mathcal{B}$. We want to show $\mathcal{C} = \mathcal{T}$

First, let $\{B_\alpha\}_{\alpha \in I}$ be a collection of elements of $\mathcal{B}$. Since $\mathcal{B} \subseteq \mathcal{T}$ and $\mathcal{T}$ is a topology, the union

$$\bigcup_{\alpha \in I} B_\alpha$$

belongs to $\mathcal{T}$, i.e. $\mathcal{C} \subseteq \mathcal{T}$.

Conversely, let $U \in \mathcal{T}$. For each $x \in U$, choose an element $B_x \in \mathcal{B}$ such that

$$x \in B_x \subseteq U.$$

Then

$$U = \bigcup_{x \in U} B_x \in \mathcal{C},$$

and hence U is a union of elements of $\mathcal{B}$, i.e. $\mathcal{T} \subseteq \mathcal{C}$.

Therefore, $\mathcal{T}$ is exactly the collection of all unions of elements of $\mathcal{B}$. □

Lemma 2.11. Let $(X, \mathcal{T})$ be a topological space. Suppose that $\mathcal{C}$ is a collection of open sets of X such that for each open set $U \subseteq X$ and each $x \in U$, there exists an element $C \in \mathcal{C}$ with

$$x \in C \subseteq U.$$

Then $\mathcal{C}$ is a basis for the topology $\mathcal{T}$ on X .

Proof. We must show that $\mathcal{C}$ is a basis. The first condition for a basis is easy: Given $x \in X$, since X is itself an open set, there is by hypothesis, an element C of $\mathcal{C}$ such that

$$x \in C \subseteq X.$$

To check the second condition, let x belong to $C_1 \cap C_2$, where C_1 and C_2 are elements of $\mathcal{C}$. Since C_1 and C_2 are open, so is $C_1 \cap C_2$. Therefore, there exists by hypothesis an element C_3 in $\mathcal{C}$ such that

$$x \in C_3 \subseteq C_1 \cap C_2.$$

Let $\mathcal{T}$ be the collection of open sets of X ; we must show that the topology $\mathcal{T}'$ generated by $\mathcal{C}$ equals the topology $\mathcal{T}$. First, note that if U belongs to $\mathcal{T}$ and if $x \in U$, then there is by hypothesis an element C of $\mathcal{C}$ such that

$$x \in C \subseteq U.$$

It follows that U belongs to the topology $\mathcal{T}'$, by definition.

Conversely, if W belongs to the topology $\mathcal{T}'$, then W equals a union of elements of $\mathcal{C}$, by the preceding lemma. Since each element of $\mathcal{C}$ belongs to $\mathcal{T}$ and $\mathcal{T}$ is a topology, W also belongs to $\mathcal{T}$. □

Lemma 2.12. *Let $\mathcal{B}$ and $\mathcal{B}'$ be bases for the topologies $\mathcal{T}$ and $\mathcal{T}'$, respectively, on X . Then the following are equivalent:*

- (1) $\mathcal{T}'$ is finer than $\mathcal{T}$.
- (2) For each $x \in X$ and each basis element $B \in \mathcal{B}$ containing x , there is a basis element $B' \in \mathcal{B}'$ such that

$$x \in B' \subseteq B.$$

Proof. (1) $\Rightarrow$ (2). We are given $x \in X$ and $B \in \mathcal{B}$, with $x \in B$. Now B belongs to $\mathcal{T}$ by definition, and $\mathcal{T} \subseteq \mathcal{T}'$ by condition (1); therefore, $B \in \mathcal{T}'$. Since $\mathcal{T}'$ is generated by $\mathcal{B}'$, there is an element $B' \in \mathcal{B}'$ such that

$$x \in B' \subseteq B.$$

(2) $\Rightarrow$ (1). Given an element U of $\mathcal{T}$, we wish to show that $U \in \mathcal{T}'$. Let $x \in U$. Since $\mathcal{B}$ generates $\mathcal{T}$, there is an element $B \in \mathcal{B}$ such that

$$x \in B \subseteq U.$$

Condition (2) tells us there exists an element $B' \in \mathcal{B}'$ such that

$$x \in B' \subseteq B.$$

Then $x \in B' \subseteq U$, so $U \in \mathcal{T}'$, by definition. □

Exercise 2.13. Let $X = \mathbb{R}^2$. Define $\mathcal{B} = \{B_r(a) : (x, y) \in \mathbb{R}^2 | (x - a_1)^2 + (y - a_2)^2 < r, \text{ where } r > 0, a = (a_1, a_2) \in \mathbb{R}^2\}$ and $\mathcal{B}' = \{B'_r(a) : (x, y) \in \mathbb{R}^2 \text{ such that } |x - a_1| < r_1 \text{ and } |y - a_2| < r_2, \text{ where } r_1, r_2 > 0, a = (a_1, a_2) \in \mathbb{R}^2\}$. Let $\mathcal{T}$ and $\mathcal{T}'$ be the topology generated by $\mathcal{B}$ and $\mathcal{B}'$ respectively. Are $\mathcal{T}$ and $\mathcal{T}'$ comparable?

Exercise 2.14. Give some examples of topologies on $\mathbb{R}$. Show that some of these topologies are comparable (one is finer or coarser than another), while some of them are not comparable.

Hint. You may consider the following topologies on $\mathbb{R}$:

- the trivial (indiscrete) topology,
- the discrete topology,
- the usual (standard) topology,
- the cofinite topology.

Determine which pairs of these topologies satisfy $\mathcal{T}_1 \subseteq \mathcal{T}_2$ or $\mathcal{T}_2 \subseteq \mathcal{T}_1$, and give examples of pairs that are not comparable.

Definition 2.15. A subbasis $\mathcal{S}$ for a topology on X is a collection of subsets of X whose union equals X . The topology generated by the subbasis $\mathcal{S}$ is defined to be the collection $\mathcal{T}$ of all unions of finite intersections of elements of $\mathcal{S}$.

Note: In the above definition, it is stated that $\mathcal{T}$ is a topology on X . However, this is not immediate and must be verified. One must check that the collection $\mathcal{T}$ satisfies the axioms of a topology. (Check!!)

2.2. The Product Topology. Let $(X, \mathcal{T}_1)$ and $(Y, \mathcal{T}_2)$ be two topological spaces. Let $X \times Y = \{(x, y) | x \in X, y \in Y\}$ be the Cartesian product of X and Y . Define $\mathcal{T}_1 \times \mathcal{T}_2 = \{U \times V | U \text{ in } \mathcal{T}_1, V \in \mathcal{T}_2\}$.

Question 2.16. Is $(X \times Y, \mathcal{T}_1 \times \mathcal{T}_2)$ a topological space?

Hint. The collection $\mathcal{T}_1 \times \mathcal{T}_2$ need not be a topology. In particular, Axiom 2 of the definition of a topology (closure under arbitrary unions) may fail. Find an explicit example where this axiom does not hold.

Claim: $\mathcal{B} = \mathcal{T}_1 \times \mathcal{T}_2$ is a basis for a topology $\mathcal{T}$.

Proof of the claim: The first condition is trivial, since $X \times Y$ is itself a basis element. To verify the second condition, let $(x, y) \in B_1 \cap B_2$, where $B_1 = U_1 \times V_1$ and $B_2 = U_2 \times V_2$, with $U_1, U_2 \in \mathcal{T}_1$ and $V_1, V_2 \in \mathcal{T}_2$. Using the identity

$$B_1 \cap B_2 = (U_1 \cap U_2) \times (V_1 \cap V_2),$$

and the fact that $\mathcal{T}_1$ and $\mathcal{T}_2$ are topologies, we have $U_1 \cap U_2 \in \mathcal{T}_1$ and $V_1 \cap V_2 \in \mathcal{T}_2$. Hence $B_1 \cap B_2 \in \mathcal{T}_1 \times \mathcal{T}_2$.

Therefore, we may take

$$B_3 = B_1 \cap B_2,$$

which contains (x, y) and satisfies $B_3 \subseteq B_1 \cap B_2$.

Definition 2.17. Let X and Y be topological spaces. The *product topology* on $X \times Y$ is the topology having as a basis the collection $\mathcal{B}$ of all sets of the form $U \times V$, where U is an open subset of X and V is an open subset of Y .

Theorem 2.18. If $\mathcal{B}_1$ is a basis for the topology of X and $\mathcal{B}_2$ is a basis for the topology of Y , then the collection

$$\mathcal{D} = \{B_1 \times B_2 \mid B_1 \in \mathcal{B}_1, B_2 \in \mathcal{B}_2\}$$

is a basis for the topology of $X \times Y$.

Proof. We apply Lemma 2.11. Given an open set W of $X \times Y$ and a point $(x, y) \in W$, by definition of the product topology, there exists a basis element $U \times V$ such that

$$(x, y) \in U \times V \subseteq W.$$

Because $\mathcal{B}_1$ and $\mathcal{B}_2$ are bases for X and Y , respectively, we can choose an element $B_1 \in \mathcal{B}_1$ such that

$$x \in B_1 \subseteq U,$$

and an element $B_2 \in \mathcal{B}_2$ such that

$$y \in B_2 \subseteq V.$$

Then

$$(x, y) \in B_1 \times B_2 \subseteq W.$$

Thus the collection $\mathcal{D}$ meets the criterion of Lemma 2.11, so $\mathcal{D}$ is a basis for $X \times Y$. $\square$

Question 2.19. Can you provide a subbasis for $(X \times Y, \mathcal{T})$?

Hint: The following theorem provides a useful tool for answering this question.

Definition 2.20. Let $\pi_1 : X \times Y \rightarrow X$ be defined by

$$\pi_1(x, y) = x,$$

and let $\pi_2 : X \times Y \rightarrow Y$ be defined by

$$\pi_2(x, y) = y.$$

The maps π_1 and π_2 are called the *projections* of $X \times Y$ onto its first and second factors, respectively.

If U is an open subset of X , then the set $\pi_1^{-1}(U)$ is precisely the set $U \times Y$, which is open in $X \times Y$. Similarly, if V is an open subset of Y , then $\pi_2^{-1}(V) = X \times V$, which is also open in $X \times Y$ with $\pi_1^{-1}(U) \cap \pi_2^{-1}(V) = U \times V$.

Theorem 2.21. *The collection*

$$\mathcal{S} = \{ \pi_1^{-1}(U) \mid U \text{ is open in } X \} \cup \{ \pi_2^{-1}(V) \mid V \text{ is open in } Y \}$$

is a subbasis for $(X \times Y, \mathcal{T})$.

Proof. Let $\mathcal{T}'$ be the topology generated by the subbasis $\mathcal{S}$. Because every element of $\mathcal{S}$ belongs to $\mathcal{T}$, it follows that arbitrary unions of finite intersections of elements of $\mathcal{S}$ also belong to $\mathcal{T}$. Hence,

$$\mathcal{T}' \subseteq \mathcal{T}.$$

On the other hand, every basis element $U \times V$ for the topology $\mathcal{T}$ is a finite intersection of elements of $\mathcal{S}$, since

$$U \times V = \pi_1^{-1}(U) \cap \pi_2^{-1}(V).$$

Therefore, $U \times V \in \mathcal{T}'$, and hence

$$\mathcal{T} \subseteq \mathcal{T}'.$$

Thus, $\mathcal{T} = \mathcal{T}'$. □

2.3. Subspace topology. Let $(X, \mathcal{T})$ be a topological space and let $Y \subseteq X$. Define $\mathcal{T}_Y = \{U \cap Y \mid U \in \mathcal{T}\}$.

Claim: $(Y, \mathcal{T}_Y)$ is topological space.

Proof of the claim: $\mathcal{T}_Y$ contains $\emptyset$ and Y , since

$$\emptyset = Y \cap \emptyset \quad \text{and} \quad Y = Y \cap X,$$

where $\emptyset$ and X are elements of $\mathcal{T}$. The fact that $\mathcal{T}_Y$ is closed under finite intersections and arbitrary unions follows from the equations

$$\bigcup_{\alpha} (U_{\alpha} \cap Y) = \left(\bigcup_{\alpha} U_{\alpha} \right) \cap Y, \quad \bigcap_{i=1}^n (U_i \cap Y) = (\bigcap_{i=1}^n U_i) \cap Y,$$

where each $U_{\alpha}, U_1, U_2 \in \mathcal{T}$. Hence $\mathcal{T}_Y$ is a topology on Y .

Definition 2.22. Let $(X, \mathcal{T})$ be a topological space. If Y is a subset of X , then $\mathcal{T}_Y$ is a topology on Y , called the *subspace topology*.

Exercise 2.23. Let $(X, \mathcal{T})$ be a topological space and let $Y \subseteq X$ be equipped with the subspace topology $\mathcal{T}_Y$. If V is an open set in the subspace $(Y, \mathcal{T}_Y)$, is V necessarily open in $(X, \mathcal{T})$? If not, give a counterexample; otherwise, prove the statement.

2.4. Closed sets.

Definition 2.24. A subset A of a topological space X is said to be *closed* if its complement $X \setminus A$ is open in X .

Example 2.25. The subset $[a, b]$ of $(\mathbb{R}, \mathcal{T}_u)$ is closed because its complement

$$\mathbb{R} \setminus [a, b] = (-\infty, a) \cup (b, +\infty)$$

is open in $\mathbb{R}$. Since $(-\infty, a) = \bigcup_{n=1}^{\infty} (-n, a)$ is open (by the 2nd axiom of definition of topology). Similarly, (b, ∞) is also open. □

Example 2.26. The interval $[a, b)$ is not closed in $\mathbb{R}$ with the usual topology. Suppose, that $[a, b)$ is closed. Then its complement

$$\mathbb{R} \setminus [a, b)$$

is open in the usual topology on $\mathbb{R}$.

Let

$$\mathbb{R} \setminus [a, b) = (-\infty, a) \cup [b, \infty).$$

Since this set is open, and $b \in [b, \infty)$, there must exist a basis element (i.e., an open interval) (c_α, d_α) such that

$$b \in (c_\alpha, d_\alpha) \subseteq \mathbb{R}.$$

But $c_\alpha \notin [b, \infty) \implies \exists x \in \mathbb{R}$ such that $x \notin [b, \infty)$ which is a contradiction that $x \in (c_\alpha, d_\alpha)$.

Therefore, $[a, b)$ is not closed in the usual topology on $\mathbb{R}$. □

Exercise 2.27. Let $(X, \mathcal{T}_d)$ be the discrete topological space. Prove that every subset of X is a closed set.

Exercise 2.28. Let $(X, \mathcal{T}_{cf})$ be a topological space endowed with the cofinite topology. Describe all the closed sets in $(X, \mathcal{T}_{cf})$.

Theorem 2.29. Let X be a topological space. Then the following conditions hold:

- (1) $\emptyset$ and X are closed.
- (2) Arbitrary intersections of closed sets are closed.
- (3) Finite unions of closed sets are closed.

Hint: Use the definition of topology and the property of the complement of a set,

Theorem 2.30. Let Y be a subspace of a topological space X . A subset $A \subseteq Y$ is closed in Y if and only if there exists a closed set $C \subseteq X$ such that

$$A = C \cap Y.$$

Proof. Assume first that $A = C \cap Y$, where C is closed in X (see Figure 17.1). Then $X \setminus C$ is open in X , and hence

$$(X \setminus C) \cap Y$$

is open in Y by definition of the subspace topology. But

$$(X \setminus C) \cap Y = Y \setminus A.$$

Therefore, $Y \setminus A$ is open in Y , and hence A is closed in Y .

Conversely, assume that A is closed in Y (see Figure 17.2). Then $Y \setminus A$ is open in Y , so by definition of the subspace topology, there exists an open set $U \subseteq X$ such that

$$Y \setminus A = U \cap Y.$$

The set $X \setminus U$ is closed in X , and we have

$$A = Y \cap (X \setminus U).$$

Thus, A is the intersection of a closed set of X with Y , as required. □

Definition 2.31. Let A be a subset of a topological space X . The *interior* of A is defined as the union of all open sets contained in A , and the *closure* of A is defined as the intersection of all closed sets containing A .

The interior of A is denoted by $\text{Int } A$, and the closure of A is denoted by $\text{Cl } A$ or by $\overline{A}$. Obviously, $\text{Int } A$ is an open set and $\overline{A}$ is a closed set; furthermore,

$$\text{Int } A \subseteq A \subseteq \overline{A}.$$

Theorem 2.32. *Let Y be a subspace of a topological space X , and let A be a subset of Y . Let $\bar{A}$ denote the closure of A in X . Then the closure of A in Y is equal to $\bar{A} \cap Y$.*

Proof. Let B denote the closure of A in Y . Since $\bar{A}$ is closed in X , the set $\bar{A} \cap Y$ is closed in Y . Moreover, B contains A . By definition, B is the intersection of all closed subsets of Y containing A ; hence,

$$B \subseteq \bar{A} \cap Y.$$

On the other hand, B is closed in Y . By Theorem 2.30 there exists a closed set $C \subseteq X$ such that $B = C \cap Y$. Since C is a closed subset of X containing A , and since $\bar{A}$ is the intersection of all closed subsets of X containing A , we have

$$\bar{A} \subseteq C.$$

Intersecting both sides with Y , we obtain

$$\bar{A} \cap Y \subseteq C \cap Y = B.$$

Combining the two inclusions, we conclude that

$$B = \bar{A} \cap Y.$$

□

□

Theorem 2.33. *Let A be a subset of the topological space X .*

- (a) *Then $x \in \bar{A}$ if and only if every open set U containing x intersects A .*
- (b) *Supposing the topology of X is given by a basis, then $x \in \bar{A}$ if and only if every basis element B containing x intersects A .*

Proof. Consider the statement in (a). It is a statement of the form $P \Rightarrow Q$. Let us transform this implication to its contrapositive, thereby obtaining the logically equivalent statement $(\text{not } Q) \Rightarrow (\text{not } P)$. Written out, it is the following:

$$x \notin \bar{A} \iff \text{there exists an open set } U \text{ containing } x \text{ that does not intersect } A.$$

In this form, the theorem is easy to prove. If $x \notin \bar{A}$, the set $U = X \setminus A$ is an open set containing x that does not intersect A , as desired. Conversely, if there exists an open set U containing x that does not intersect A , then $X \setminus U$ is a closed set containing A . By definition of the closure $\bar{A}$, the set $X \setminus U$ must contain $\bar{A}$; therefore, $x \notin \bar{A}$.

Statement (b) follows readily. If every open set containing x intersects A , so does every basis element B containing x , because B is an open set. Conversely, if every basis element containing x intersects A , so does every open set U containing x , because U contains a basis element that contains x . □

Definition 2.34. Let A be a subset of a topological space X , and let $x \in X$. We say that x is a *limit point* (or *cluster point*, or *point of accumulation*) of A if every neighborhood of x intersects A in some point other than x itself.

Example 2.35. Consider the real line $\mathbb{R}$. Let

$$A = (0, 1].$$

Then the point 0 is a limit point of A , and so is the point 1. In fact, every point of the interval $[0, 1]$ is a limit point of A , while no point of $\mathbb{R} \setminus [0, 1]$ is a limit point of A . □

Theorem 2.36. *Let A be a subset of a topological space X , and let A' denote the set of all limit points of A . Then*

$$\bar{A} = A \cup A'.$$

2.5. Hausdorff Space.

Definition 2.37. Let X be a topological space. A sequence $x_1, x_2, \dots$ of points of X is said to *converge* to a point $x \in X$ if, corresponding to each neighborhood U of x , there exists a positive integer N such that

$$x_n \in U \quad \text{for all } n \geq N.$$

In $\mathbb{R}$ and $\mathbb{R}^2$, a sequence cannot converge to more than one point. However, in an arbitrary topological space this need not be the case. For example, let $X = \{a, b, c\}$, $\mathcal{T} = \{\emptyset, X, \{b\}, \{a, b\}, \{b, c\}\}$ be a topological space and the sequence defined by

$$x_n = b \quad \text{for all } n$$

converges not only to the point b , but also to the points a and c .

Definition 2.38. A topological space X is called a *Hausdorff space* if for each pair x_1, x_2 of distinct points of X , there exist neighborhoods U_1 of x_1 and U_2 of x_2 , respectively, such that

$$U_1 \cap U_2 = \emptyset.$$

Theorem 2.39. *Every finite point set in a Hausdorff space X is closed.*

Proof. It suffices to show that every one-point set $\{x_0\}$ is closed. Let $x \in X$ be a point different from x_0 . Since X is Hausdorff, there exist disjoint neighborhoods U of x and V of x_0 .

Because $U \cap \{x_0\} = \emptyset$, the point x cannot belong to the closure of the set $\{x_0\}$. Therefore, no point other than x_0 lies in the closure of $\{x_0\}$.

As a result, the closure of the set $\{x_0\}$ is $\{x_0\}$ itself, and hence $\{x_0\}$ is closed. $\square$

Question 2.40. *In a topological space X , suppose that every finite subset of X is closed. Does it follow that X is Hausdorff?*

Answer: No. The following is a counterexample.

Let X be an infinite set equipped with the *cofinite topology*

$$\mathcal{T}_{cf} = \{\emptyset\} \cup \{U \subseteq X \mid X \setminus U \text{ is finite}\}.$$

- Every finite subset $F \subseteq X$ is closed, since $X \setminus F$ is cofinite and therefore open.
- The space $(X, \mathcal{T}_{cf})$ is not Hausdorff. Indeed, if $x \neq y$ are points of X , then any open sets U containing x and V containing y have finite complements. Hence,

$$U \cap V \neq \emptyset,$$

so x and y cannot be separated by disjoint open neighborhoods.

Conclusion. Although every finite subset of X is closed, the space need not be Hausdorff. Therefore,

$$\text{every finite set is closed} \not\Rightarrow X \text{ is Hausdorff.}$$

Definition 2.41. A topological space X is said to satisfy the T_1 -axiom if every finite set is closed.

Theorem 2.42. *Let X be a topological space satisfying the T_1 axiom, and let $A \subseteq X$. Then a point $x \in X$ is a limit point of A if and only if every neighborhood of x contains infinitely many points of A .*

Proof. If every neighborhood of x intersects A in infinitely many points, then it certainly intersects A in some point other than x itself. Hence, x is a limit point of A .

Conversely, suppose that x is a limit point of A , and assume that there exists a neighborhood U of x that intersects A in only finitely many points. Then U also intersects $A \setminus \{x\}$ in finitely many points; let

$$\{x_1, x_2, \dots, x_m\}$$

be the points of $U \cap (A \setminus \{x\})$.

Since X satisfies the T_1 axiom, the finite point set $\{x_1, x_2, \dots, x_m\}$ is closed. Therefore, the set

$$X \setminus \{x_1, x_2, \dots, x_m\}$$

is open in X . Hence,

$$U \cap (X \setminus \{x_1, x_2, \dots, x_m\})$$

is a neighborhood of x that does not intersect $A \setminus \{x\}$ at all. This contradicts the assumption that x is a limit point of A .

Therefore, every neighborhood of a limit point x of A must intersect A in infinitely many points. □

Theorem 2.43. *If X is a Hausdorff space, then a sequence of points of X converges to at most one point of X .*

Proof. Suppose that (x_n) is a sequence of points of X that converges to x . Let $y \in X$ with $y \neq x$. Since X is Hausdorff, there exist disjoint neighborhoods U of x and V of y .

Because $x_n \rightarrow x$, the neighborhood U contains x_n for all but finitely many values of n . Hence, for sufficiently large n ,

$$x_n \notin V,$$

since $U \cap V = \emptyset$. Therefore, the sequence (x_n) cannot eventually lie in V , and thus cannot converge to y .

Hence, x_n converges to at most one point. □

2.6. Continuous Functions.

Definition 2.44. Let X and Y be topological spaces. A function $f : X \rightarrow Y$ is said to be *continuous* if for each open subset $V \subseteq Y$, the set $f^{-1}(V)$ is an open subset of X .

Theorem 2.45. *Let X and Y be topological spaces, and let $f : X \rightarrow Y$. Then the following are equivalent:*

- (1) f is continuous.
- (2) For every subset $A \subseteq X$, one has $f(\overline{A}) \subseteq \overline{f(A)}$.
- (3) For every closed set $B \subseteq Y$, the set $f^{-1}(B)$ is closed in X .
- (4) For each $x \in X$ and each neighborhood V of $f(x)$, there is a neighborhood U of x such that

$$f(U) \subseteq V.$$

Proof. We show that (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (1) and that (1) $\Leftrightarrow$ (4).

(1) $\Rightarrow$ (2). Assume that f is continuous. Let A be a subset of X . We show that if $x \in \overline{A}$, then $f(x) \in \overline{f(A)}$. Let V be a neighborhood of $f(x)$. Then $f^{-1}(V)$ is an open set of X containing x ; hence it must intersect A in some point y . Then V intersects $f(A)$ in the point $f(y)$, so that $f(x) \in \overline{f(A)}$, as desired.

(2) $\Rightarrow$ (3). Let B be closed in Y and let $A = f^{-1}(B)$. We wish to prove that A is closed in X ; that is, we show that $\overline{A} = A$. By elementary set theory, we have

$$f(A) = f(f^{-1}(B)) \subseteq B.$$

Therefore, if $x \in \overline{A}$, then

$$f(x) \in \overline{f(A)} \subseteq \overline{B} = B,$$

so that $x \in f^{-1}(B) = A$. Thus $\overline{A} \subseteq A$, and hence $A = \overline{A}$, as desired.

(3) $\Rightarrow$ (1). Let V be an open set of Y . Set $B = Y \setminus V$. Then

$$f^{-1}(B) = f^{-1}(Y) \setminus f^{-1}(V) = X \setminus f^{-1}(V).$$

Since B is closed in Y , the set $f^{-1}(B)$ is closed in X by hypothesis. Therefore, $f^{-1}(V)$ is open in X , and f is continuous.

(1) $\Rightarrow$ (4). Assume that f is continuous. Let $x \in X$, and let V be a neighborhood of $f(x)$. Then $f^{-1}(V)$ is an open neighborhood of x . Let $U = f^{-1}(V)$. Then $f(U) \subseteq V$, as required.

(4) $\Rightarrow$ (1). Let V be an open set of Y . For each $x \in f^{-1}(V)$, the set V is a neighborhood of $f(x)$. By hypothesis, there exists a neighborhood U_x of x such that $f(U_x) \subseteq V$. Then

$$f^{-1}(V) = \bigcup_{x \in f^{-1}(V)} U_x,$$

which is a union of open sets and hence open. Therefore, f is continuous. □

Example 2.46. Let X be a nonempty set with $|X| \geq 2$. Let $\mathcal{T}_{ind} = \{\emptyset, X\}$ be the *indiscrete topology* on X , and let $\mathcal{T}_d = \mathcal{P}(X)$ be the *discrete topology* on X .

Consider the identity map

$$i : (X, \mathcal{T}_{ind}) \longrightarrow (X, \mathcal{T}_d), \quad i(x) = x.$$

This map is *not continuous*. Indeed, every singleton set $\{x\}$ is open in the discrete topology $\mathcal{T}_{dis}$, but

$$i^{-1}(\{x\}) = \{x\}$$

is not open in the indiscrete topology $\mathcal{T}_{ind}$, since the only open sets in $\mathcal{T}_{ind}$ are $\emptyset$ and X .

Hence, the identity map from the indiscrete topology to the discrete topology fails to be continuous. □

Example 2.47. Let X be a set equipped with the discrete topology $\mathcal{T}_{dis}$, and let Y be any topological space. Let

$$f : (X, \mathcal{T}_{dis}) \rightarrow Y$$

be any function.

Then f is continuous. Indeed, if $V \subseteq Y$ is open, then

$$f^{-1}(V) \subseteq X.$$

Since every subset of X is open in the discrete topology, $f^{-1}(V)$ is open in $(X, \mathcal{T}_{dis})$. Hence, f is continuous. □

Definition 2.48. Let X and Y be topological spaces. The function f is called a *homeomorphism* if the following conditions hold:

- (1) $f : X \rightarrow Y$ is bijective.
- (2) f is continuous.
- (3) The inverse function $f^{-1} : Y \rightarrow X$ is continuous.

Theorem 2.49. Let X and Y be topological spaces, and let $f : X \rightarrow Y$ be a bijection. Then f is a homeomorphism if and only if for every subset $U \subseteq X$,

$$U \text{ is open in } X \iff f(U) \text{ is open in } Y.$$

Proof. ($\Rightarrow$) Assume that $f : X \rightarrow Y$ is a homeomorphism. Then f is bijective, continuous, and f^{-1} is continuous. Let $U \subseteq X$.

If U is open in X , then by continuity of f^{-1} ,

$$f(U) = (f^{-1})^{-1}(U)$$

is open in Y . Conversely, if $f(U)$ is open in Y , then by continuity of f ,

$$U = f^{-1}(f(U))$$

is open in X . Hence,

$$U \text{ is open in } X \iff f(U) \text{ is open in } Y.$$

($\Leftarrow$) Conversely, if $f : X \rightarrow Y$ is a bijection such that U is open in X if and only if $f(U)$ is open in Y , then both f and f^{-1} preserve open sets, so both are continuous. Hence f is a homeomorphism. $\square$

Theorem 2.50. (The pasting lemma) Let $X = A \cup B$, where A and B are closed subsets of X . Let $f : A \rightarrow Y$ and $g : B \rightarrow Y$ be continuous functions. If $f(x) = g(x)$ for every $x \in A \cap B$, then f and g combine to give a continuous function $h : X \rightarrow Y$, defined by

$$h(x) = \begin{cases} f(x), & \text{if } x \in A, \\ g(x), & \text{if } x \in B. \end{cases}$$

Proof. Let C be a closed subset of Y . By elementary set theory, we have

$$h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C).$$

Since f is continuous, the set $f^{-1}(C)$ is closed in A , and hence closed in X because A is closed in X . Similarly, since g is continuous, the set $g^{-1}(C)$ is closed in B , and therefore closed in X .

Thus, $h^{-1}(C)$, being the union of two closed subsets of X , is closed in X . Hence, h is continuous. $\square$

2.7. The Metric Topology.

Definition 2.51. A *metric* on a set X is a function

$$d : X \times X \rightarrow \mathbb{R}$$

satisfying the following properties for all $x, y, z \in X$:

- (1) **(Positivity)** $d(x, y) \geq 0$, with equality if and only if $x = y$.
- (2) **(Symmetry)** $d(x, y) = d(y, x)$.
- (3) **(Triangle inequality)** $d(x, y) + d(y, z) \geq d(x, z)$.

Definition 2.52. Given a metric d on a set X , the number $d(x, y)$ is called the *distance* between x and y . Given $\varepsilon > 0$, consider the set

$$B_d(x, \varepsilon) = \{y \in X \mid d(x, y) < \varepsilon\},$$

consisting of all points whose distance from x is less than ε . This set is called the ε -ball centered at x .

When no confusion can arise, we omit the metric d from the notation and write the ball simply as $B(x, \varepsilon)$.

Lemma 2.53. Let (X, d) be a metric space. Each $y \in B_d(x, \epsilon)$, there exists $\delta > 0$ such that $B_d(y, \delta) \subseteq B_d(x, \epsilon)$.

Proof. Since $y \in B_d(x, \varepsilon)$, we have $d(x, y) < \varepsilon$. Set $\delta = \varepsilon - d(x, y) > 0$. Let $z \in B_d(y, \delta)$. Then

$$d(y, z) < \delta.$$

By the triangle inequality,

$$d(x, z) \leq d(x, y) + d(y, z) < d(x, y) + \delta = \varepsilon.$$

Hence $z \in B_d(x, \varepsilon)$, and therefore

$$B_d(y, \delta) \subseteq B_d(x, \varepsilon).$$

□

Definition-Proposition 1. If d is a metric on a set X , then the collection of all ε -balls $\{B_d(x, \varepsilon) | x \in X, \varepsilon > 0\}$, forms a basis for a topology on X . This topology is called the *metric topology* induced by d .

Proof. Let

$$\mathcal{B} = \{B_d(x, \varepsilon) \mid x \in X, \varepsilon > 0\}.$$

We show that $\mathcal{B}$ is a basis for a topology on X .

(1) *Covering condition.* For each $x \in X$ and any $\varepsilon > 0$, we have

$$x \in B_d(x, \varepsilon),$$

so every point of X lies in some element of $\mathcal{B}$.

(2) *Intersection condition.* Let $B_d(x, \varepsilon)$ and $B_d(x', \varepsilon')$ be two basis elements, and let

$$y \in B_d(x, \varepsilon) \cap B_d(x', \varepsilon').$$

By the previous result, there exists $\delta_1 > 0$ such that

$$B_d(y, \delta_1) \subseteq B_d(x, \varepsilon),$$

and there exists $\delta_2 > 0$ such that

$$B_d(y, \delta_2) \subseteq B_d(x', \varepsilon').$$

Let

$$\delta = \min\{\delta_1, \delta_2\}.$$

Then

$$B_d(y, \delta) \subseteq B_d(x, \varepsilon) \cap B_d(x', \varepsilon'),$$

and $B_d(y, \delta) \in \mathcal{B}$.

Since $\mathcal{B}$ satisfies both basis axioms, it is a basis for a topology on X . This topology is called the *metric topology* induced by d . □

Example 2.54. The standard metric on the real numbers $\mathbb{R}$ is defined by

$$d(x, y) = |x - y|.$$

□

Example 2.55. Given a set X , define a function $d : X \times X \rightarrow \mathbb{R}$ by

$$d(x, y) = \begin{cases} 1, & \text{if } x \neq y, \\ 0, & \text{if } x = y. \end{cases}$$

It is straightforward to verify that d is a metric on X . The topology induced by this metric is the *discrete topology*. For example, the basis element $B_d(x, 1)$ consists of the single point x . □

REFERENCES

- [1] James R. Munkres, *Topology*, 2nd edition, Pearson Education, 2000.
- [2] Colin Adams and Robert Franzosa, *Introduction to Topology, pure and applied*, 1st edition, Pearson Education, 2008.

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